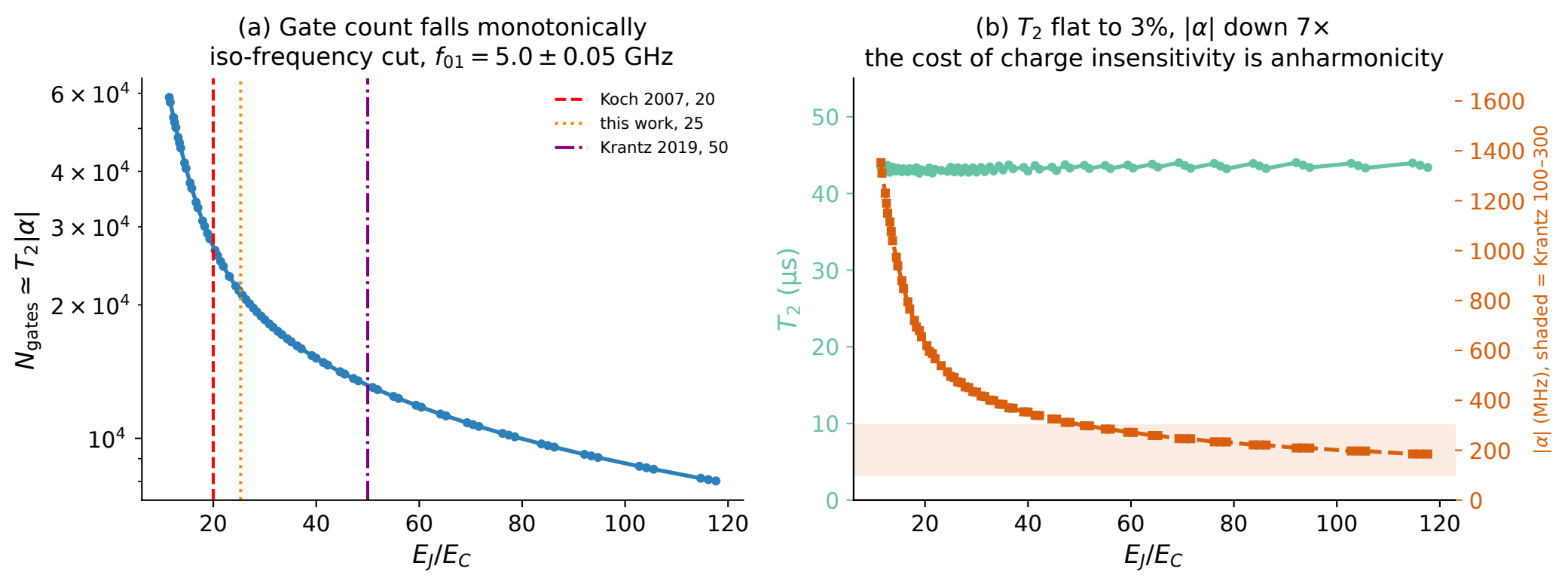

(a) Gate count falls monotonically
iso-frequency cut, $f_{01} = 5.0 \pm 0.05$ GHz
$N_{gates} \approx T_2|\alpha|$
6×10^4
4×10^4
3×10^4
2×10^4
10^4
Koch 2007, 20
this work, 25
Krantz 2019, 50
20
40
60
80
100
120
E_J/E_C
(b) T_2 flat to 3%, $|\alpha|$ down 7×
the cost of charge insensitivity is anharmonicity
T_2 (μs)
50
40
30
20
10
0
1600
1400
1200
1000
800
600
400
200
0
$|\alpha|$ (MHz), shaded = Krantz 100–300
20
40
60
80
100
120
E_J/E_C